

Commerce XI Maths App

Exercise 2.3

Detailed Answers with Explanations and Steps

Mathematics & Statistics, Part 2, Unit 2 - Measures of Dispersion | Total questions: 16

Question 1 - Q1

Mean and standard deviation of two distributions of 100 and 150 items are 50, 5 and 40, 6 respectively. Find the mean and standard deviations of all the 250 items taken together.

Options

A. Mean = 44, S. D. = $\sqrt{55.6} \approx 7.46$

B. Mean = 45, S. D. = $\sqrt{55.6} \approx 7.46$

C. Mean = 44, S. D. = 55.6

D. Mean = 40, S. D. = $\sqrt{49} = 7$

Correct answer: Mean = 44, S. D. = $\sqrt{55.6} \approx 7.46$

Explanation and steps

Combined mean = $(100 \times 50 + 150 \times 40)/250 = 44$. Combined variance = $[100\{25 + (50-44)^2\} + 150\{36 + (40-44)^2\}]/250 = 55.6$. Therefore combined S.D. = $\sqrt{55.6}$, approximately 7.46. Textbook answer note: the extracted/printed key appears as '55.6516'; it joins the variance 55.6 and its square-root value 7.46. It must not be read as $\sqrt{55.6516}$.

Question 2 - Q2

For a certain bivariate data, following information is available.

	X	Y
Mean	13	17
S.D.	3	2
Size	10	10

Obtain the combined standard deviation.

Options

A. $\sqrt{10.5}$

B. 10.5

C. $\sqrt{8.5}$

D. $\sqrt{12.5}$

Correct answer: $\sqrt{10.5}$

Explanation and steps

First find the size-weighted combined mean, then include both within-group variance and the squared difference of each group mean from the combined mean. This gives $\sqrt{10.5}$.

Question 3 - Q3

Calculate coefficient of variation of marks secured by a student in the exam, where the marks are: 2, 4, 6, 8, 10. (Given: $\sqrt{2} = 1.41$)

Correct answer: 47%

Explanation and steps

Use C.V. = $(\text{S.D.}/\text{mean}) \times 100$ and compare the percentages when required. The verified result is 47%.

Question 4 - Q4

Find the coefficient of variation of a sample which has mean 25 and standard deviation 5.

Correct answer: 20%

Explanation and steps

Use $C.V. = (S.D./\text{mean}) \times 100$ and compare the percentages when required. The verified result is 20%.

Question 5 - Q5

A group of 65 students of class XI have their average height 150.4 cm with coefficient of variation 2.5%. What is the standard deviation of their height?

Correct answer: 3.76

Explanation and steps

Use $C.V. = (S.D./\text{mean}) \times 100$ and compare the percentages when required. The verified result is 3.76.

Question 6 - Q6 · Concept

Two workers on the same job show the following results:

	Worker P	Worker Q
Mean time for completing the job (hours)	33	21
Standard Deviation (hours)	9	7

To find out which worker is more consistent, you will calculate ----, a smaller ----- means greater consistency.

Options

- A. Coefficients of Variation (C. V.)
- B. Mean
- C. Median
- D. Standard Deviation (S. D.)

Correct answer: Coefficients of Variation (C. V.)

Explanation and steps

Apply the appropriate variance formula and take its positive square root when S.D. is requested. The verified result is Coefficients of Variation (C. V.).

Question 7 - Q6 · Concept

Two workers on the same job show the following results:

	Worker P	Worker Q
Mean time for completing the job (hours)	33	21
Standard Deviation (hours)	9	7

To find out which worker seems to be faster, you will calculate-----, a smaller ---- indicates faster work.

Options

- A. Coefficients of Variation (C. V.)
- B. Mean
- C. Median
- D. Standard Deviation (S. D.)

Correct answer: Mean

Explanation and steps

Apply the appropriate variance formula and take its positive square root when S.D. is requested. The verified result is Mean.

Question 8 - Q6(i)

Two workers on the same job show the following results:

	Worker P	Worker Q
Mean time for completing the job (hours)	33	21
Standard Deviation (hours)	9	7

Regarding the time required to complete the job, which worker is more consistent?

Options

- A. Worker P
- B. Worker Q
- C. Both are equally consistent
- D. Consistency cannot be compared

Correct answer: Worker P

Explanation and steps

Apply the appropriate variance formula and take its positive square root when S.D. is requested. The verified result is Worker P.

Question 9 - Q6(ii)

Two workers on the same job show the following results:

	Worker P	Worker Q
Mean time for completing the job (hours)	33	21
Standard Deviation (hours)	9	7

Which worker seems to be faster in completing the job?

Options

- A. Worker Q
- B. Worker P
- C. Both take the same time
- D. Speed cannot be compared

Correct answer: Worker Q

Explanation and steps

Apply the appropriate variance formula and take its positive square root when S.D. is requested. The verified result is Worker Q.

Question 10 - Q7(i)

A company has two departments with 42 and 60 employees respectively.

Their average weekly wages are ₹ 750 and ₹ 400.

The standard deviations are 8 and 10 respectively.

Which department has a larger bill?

Options

- A. First department
- B. Second department
- C. Both have equal bills
- D. Cannot be determined

Correct answer: First department

Explanation and steps

Apply the appropriate variance formula and take its positive square root when S.D. is requested. The verified result is First department.

Question 11 - Q7(ii)

A company has two departments with 42 and 60 employees respectively. Their average weekly wages are ₹ 750 and ₹ 400. The standard deviations are 8 and 10 respectively. Which department has larger variability in wages?

Options

- A. Second department
- B. First department
- C. Both have equal variability
- D. Cannot be determined

Correct answer: Second department

Explanation and steps

Apply the appropriate variance formula and take its positive square root when S.D. is requested. The verified result is Second department.

Question 12 - Q8

The following table gives weights of the students of class A. Calculate the coefficient of variation (Given : $\sqrt{0.8} = 0.8944$)

Weight (in kg)	25-35	35-45	45-55
Class A	8	4	8

Correct answer: 22.36%

Explanation and steps

Use C.V. = (S.D./mean) x 100 and compare the percentages when required. The verified result is 22.36%.

Question 13 - Q9(i)

No. of goals	0	1	2	3	4
No. of matches played by Team A	19	6	5	16	14
No. of matches played by Team B	16	16	5	18	15

Compute the coefficients of variation for Team A and Team B.

Options

- A. CV(A) = 80% and CV(B) = 75%
- B. CV(A) = 75% and CV(B) = 80%
- C. CV(A) = 80% and CV(B) = 80%
- D. CV(A) = 19% and CV(B) = 16%

Correct answer: CV(A) = 80% and CV(B) = 75%

Explanation and steps

Use C.V. = (S.D./mean) x 100 and compare the percentages when required. The verified result is CV(A) = 80% and CV(B) = 75%.

Question 14 - Q9(ii)

No. of goals	0	1	2	3	4
No. of matches played by Team A	19	6	5	16	14
No. of matches played by Team B	16	16	5	18	15

Which team is more consistent?

Options

- A. Team B
- B. Team A
- C. Both are equally consistent
- D. Neither team

Correct answer: Team B

Explanation and steps

Apply the appropriate variance formula and take its positive square root when S.D. is requested. The verified result is Team B.

Question 15 - Q10(i)

Given below is the information about marks obtained in Mathematics and Statistics by 100 students in a class.

	Mathematics	Statistics
Mean	20	25
S.D.	2	3

Compute the coefficients of variation for Mathematics and Statistics.

Options

- A. $CV(M) = 10\%$ and $CV(S) = 12\%$
- B. $CV(M) = 12\%$ and $CV(S) = 10\%$
- C. $CV(M) = 20\%$ and $CV(S) = 25\%$
- D. $CV(M) = 2\%$ and $CV(S) = 3\%$

Correct answer: $CV(M) = 10\%$ and $CV(S) = 12\%$

Explanation and steps

Use $C.V. = (S.D./\text{mean}) \times 100$ and compare the percentages when required. The verified result is $CV(M) = 10\%$ and $CV(S) = 12\%$.

Question 16 - Q10(ii)

Given below is the information about marks obtained in Mathematics and Statistics by 100 students in a class.

	Mathematics	Statistics
Mean	20	25
S.D.	2	3

Which subject shows the highest variability in marks?

Options

- A. Statistics
- B. Mathematics
- C. Both have equal variability
- D. Neither subject

Correct answer: Statistics

Explanation and steps

Apply the appropriate variance formula and take its positive square root when S.D. is requested. The verified result is Statistics.